

Eclogite Facies Metamorphic Reaction

Grossular Composition Analysis

Q.9: $3 \text{ Anorthite} = \text{Grossular} + 2 \text{ Kyanite} + \text{Quartz}$ is a reaction which marks the eclogite facies in metabasic rocks. What will be the composition of grossular in the reaction given above?

(a) $\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12}$

(b) CaSiO_3

(c) $(\text{Ca}, \text{Mg})\text{Si}_2\text{O}_6$

(d) CaCO_3

✓ CORRECT ANSWER: (a) $\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12}$

1. TEXTBOOK REFERENCE & COMPOSITION OF GROSSULAR

According to Perkins (2014), grossular is the aluminum end-member of the ugrandite series of garnets. The definitive composition is explicitly stated as:

"Grossular, $\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12}$, has $\text{CaO}:\text{Al}_2\text{O}_3:\text{SiO}_2$ ratio 3:1:3, equivalent to 43% CaO, 14% Al_2O_3 , and 43% SiO_2 ." (Perkins, 2014, p. 107)

2. CHEMICAL VERIFICATION OF THE REACTION

To confirm that option (a) is correct, we can verify the stoichiometry of the given reaction:

Phase	Chemical Formula	Coefficient
Anorthite (plagioclase)	$\text{CaAl}_2\text{Si}_2\text{O}_8$	3×
↓		
Grossular (garnet)	$\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12}$	1×
Kyanite (aluminosilicate)	Al_2SiO_5	2×
Quartz (silica)	SiO_2	1×

Left side (Reactant):

$$3 \times (\text{CaAl}_2\text{Si}_2\text{O}_8) = \text{Ca}_3\text{Al}_6\text{Si}_6\text{O}_{24}$$

Right side (Products):

$$\begin{aligned} &\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12} + 2(\text{Al}_2\text{SiO}_5) + \text{SiO}_2 \\ &= \text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12} + \text{Al}_4\text{Si}_2\text{O}_{10} + \text{SiO}_2 \end{aligned}$$



✓ The equation is perfectly balanced, confirming the composition of grossular as $\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12}$

3. GARNET CLASSIFICATION & GROSSULAR

Perkins (2014) classifies garnets into two series based on their chemical composition:

"Mineralogists conveniently divide garnets into two series, the pyralspites (pyrope-almandine-spessartine) and the ugrandites (uvarovite-grossular-andradite)." (Perkins, 2014, p. 360)

Series	Composition Formula	End Members
Pyralspites	$(\text{Mg,Fe,Mn})_3\text{Al}_2\text{Si}_3\text{O}_{12}$	Pyrope, Almandine, Spessartine
Ugrandites	$\text{Ca}_3(\text{Cr,Al,Fe})_2\text{Si}_3\text{O}_{12}$	Uvarovite, Grossular, Andradite

"Grossular is the Al end member of the ugrandite series, $\text{Ca}_3[\text{Fe}^{3+},\text{Al}^{3+},\text{Cr}^{3+}]_2\text{Si}_3\text{O}_{12}$, but natural grossular may contain appreciable amounts of Fe^{3+} or Cr^{3+} in solid solution." (Perkins, 2014, p. 364)

4. ECLOGITE FACIES CONTEXT

This reaction is indeed diagnostic of the eclogite facies in metabasic (mafic) rocks. According to the textbook:

"Eclogites are mafic rocks metamorphosed at high pressure and moderate-to-high temperature. They contain the essential minerals pyrope (Mg-rich garnet) and the green Na-rich clinopyroxene called omphacite. Orthopyroxene may also be present in significant quantities. Accessory minerals include kyanite, quartz, spinel, titanite, and many others." (Perkins, 2014, p. 204)

The appearance of kyanite (high-pressure polymorph of Al_2SiO_5) along with garnet and quartz is characteristic of high-pressure metamorphic conditions. Grossular can appear in eclogite-facies rocks when metabasic protoliths are rich in aluminum and calcium-bearing plagioclase.

5. OPTION ANALYSIS

Option	Mineral/Phase	Reasoning
(a) $\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12}$	✓ Grossular Garnet	Correct composition of the aluminum end-member of the ugrandite garnet series. Chemically balances the reaction equation perfectly.
(b) CaSiO_3	Wollastonite	A calcium silicate but wrong composition. This mineral forms in lower-pressure marbles, not in the eclogite facies. Lacks aluminum.
(c) $(\text{Ca,Mg})\text{Si}_2\text{O}_6$	Diopside (pyroxene)	A calcium-magnesium silicate. While found in eclogites, it does not balance the reaction equation. Also lacks sufficient aluminum.
(d) CaCO_3	Calcite	A carbonate mineral with no silicate or aluminum content. Chemically unrelated to the anorthite reactant or the products of this metamorphic reaction.

6. SUMMARY

The composition of grossular in this eclogite facies reaction is $\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12}$, which is the pure end-member composition of the grossular garnet. This mineral represents one of the garnet phases stable at high pressure and temperature in metamorphic rocks derived from aluminum-rich mafic protoliths.

The reaction demonstrates the transformation of plagioclase-rich igneous rocks (anorthite) into high-pressure metamorphic assemblages dominated by garnet (grossular), the high-pressure aluminosilicate polymorph

(kyanite), and excess silica (quartz), which is characteristic of eclogite-facies metamorphism.

Source: Perkins, D. (2014). *Mineralogy* (3rd ed.). Pearson Education Limited.

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